
Sound change, phonological theory, and *inference*

Deepthi Gopal, Tromsø/Uppsala

§ 1

Preliminaries

What is(n't) a sound change? What is(n't) phonological change?

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 - Swedish /i/ $\neq$ anyone else's /i/, etc. Maybe even textbook surface variation in French: $r > \text{ʀ}$, ʁ , or ʁ .

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A few Scandinavian verbs.

Central Swedish

Scanian Swedish

/hø:ra/ [2'hœ:ra]

‘to hear’

/çø:ra/ [2'çœ:ra]

‘to drive’

/fø:ra/ [2'fœ:ra]

‘to bring, to take’

/lɛ:ra/ [2'læ:ra]

‘to learn’

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- **So one thing we want to know is whether or not change in the *realisation* of x has led to change in the *system* in which it participates.** Note the implied order of events: change on the surface, then change underneath. We'll return to this point; will all 'surface change' inevitably give rise to phonological change *given sufficient time*?

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Representational change without phonetic change? S-K argues that there was no phonetic 'weakening' here.

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W. Germanic stressed *a before back vowels: $a > \text{æ} > a > \text{æ} > \text{æ}\text{ə}$? (Dresher, 2015)

Early WG **habuc* 'hawk' > **hæbuc* ('Anglo-Frisian Brightening') > *hafuc* ('a-restoration') > **hæfuc* ('Second fronting') > *hæɤfuc* > *<heafuc>* in the Mercian *Vespasian Psalter*.

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a-restoration was a rule, not a $\triangleright$, which was then lost. Change in the grammar.

$$\underset{[+stressed]}{\text{æ}} \longrightarrow [+back] / ____ C \underset{[+back]}{V}$$

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<i>uninflected form</i>	<i>absolutive sg.</i>	
<i>gixon</i>	<i>gixona</i>	'man'
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- Actually probably a sequence of quite a few changes: *neskaa* > *neskea* > *neskia* > *neskie* > *neski*, with **all the intermediate forms attested in other Basque dialects**. But almost certainly silly to put it through all that in the synchronic derivation; maybe **a-a → i / _# !**

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 - In particular, how we (synchronic phonologists? diachronic phonologists?) reason from 1. global typology, 2. 'local' typology, 'dialect data', and the 'relatedness' of varieties; what we can really infer; and how right we are to do it.

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Looking only at changes that are regular in this way, as the Neogrammarians would have, we then have *unconditioned* and *conditioned* changes. Individual conditioned changes can often look suspiciously like something else we are interested in, which is to say, synchronic phonological alternations, modulo worries about reanalysis and rule inversion; but it is somewhere to start!

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But for most practical purposes, a theory of *possible / probable* changes constrains what we ought to be reconstructing. What constrains it in turn?

What's where when why?

In talking about these two varieties of Swedish, I've been assuming a few things: this is that we know what the *ancestral state* is, and that we know that these two varieties share a common origin. Easily done here, or is it? **What about when we don't know that for sure?** Our inferences about the deeper past are usually *predicated* on the assumption of regularity, the bedrock of 'the' **Comparative Method**, which lets us both establish relatedness and reconstruct states for which we don't have direct attestation.

- (Of course, there's no such thing (Meillet, 1954; Baxter, 2002); we can trace the idea of using regular correspondences to prove that lgs. are related back through a long history, but it's not the case that everyone agrees on exactly how we should do it.)
- Of course, the CM is not the only source of evidence we have for the phonology of states of languages to which we don't have direct access, on which more later.

But for most practical purposes, a theory of *possible* / *probable* changes constrains what we ought to be reconstructing. What constrains it in turn? **What assumptions have I made in saying these sentences?**

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 - And see e.g. Kümmel (2012), who points out (Blust, 1974, 2006) that Kelabit (Northern Borneo) has voiced aspirates without voiceless ones. i.e. enhance our hypothesis-formation via better *global typological attestation*.

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So in some sense, the Indo-European stop problem is about a conflict between two loci of *realism*. How do we weigh up (synchronic? uniformitarian?) **typological** concerns (do we want a language that 'looks normal'?) against **diachronic** concerns (do we want a language that (we think) is in a configuration that *increases the probability of certain changes taking place afterwards*)?

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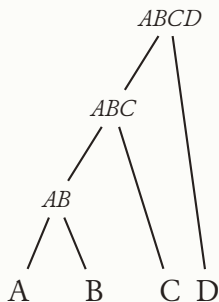
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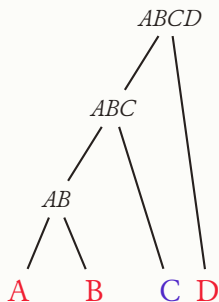
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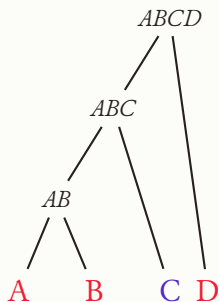
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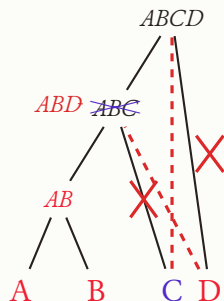
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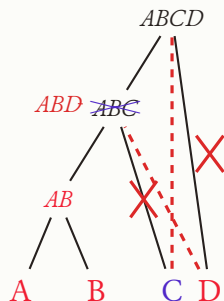
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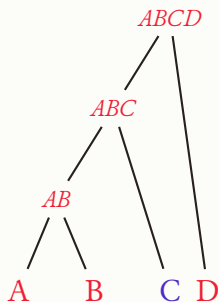
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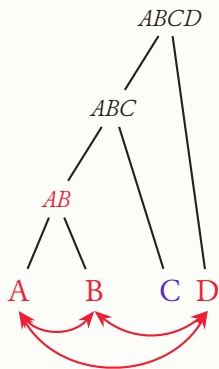
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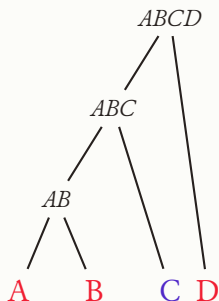
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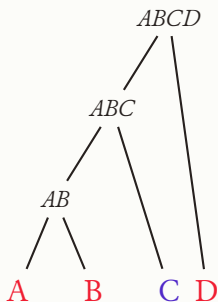
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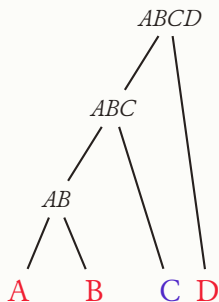
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Disambiguating between these options is, I'd argue, a job for a good theory of what phonological change can do.

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- The only difference between CSw. and HSw. is likely ± a *rule*.

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What's where when why?

Let's take a look at that Scandinavian retroflexion again.

Swedish retroflexion.

	<i>Central Swedish</i>		<i>Scanian Swedish</i>		<i>Helsinki Swedish</i>	
'hear'±P.PTCP	hø:ra	hø:ɹ̥	hø:ɤa	hø:ɤt	hø:ra	hø:rt
'drive'±P.PTCP	ɕø:ra	ɕø:ɹ̥	ɕø:ɤa	ɕø:ɤt	ɕø:ra	ɕø:rt
'bring, take'±P.PTCP	fø:ra	fø:ɹ̥	fø:ɤa	fø:ɤt	fø:ra	fø:rt
'learn'±P.PTCP	lɛ:ra	lɛ:ɹ̥	lɛ:ɤa	lɛ:ɤt	lɛ:ra	lɛ:rt

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- Is there a pattern which is logically possible but not attested? **Is that gap systematic?**

What's where when why?

Norwegian retroflexion (Stausland Johnsen, 2012).

	<i>Standard E. Norwegian</i>	<i>(Stavanger)</i>	<i>Frogner, Arendal & Tvedestrand</i>
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'big'±NEUTER	stu:r	stu:ɹ	
'father'±POSS	fa:r	fa:ɹ	

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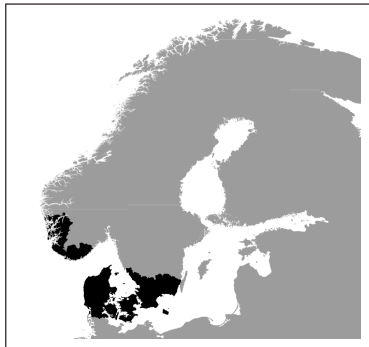
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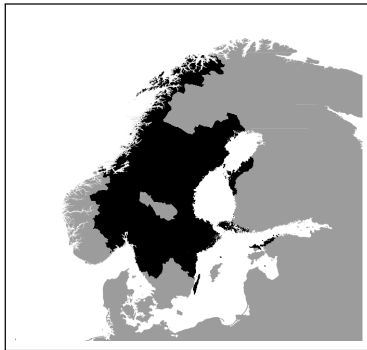
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(B) Uvular *r*



(c) Retroflexion

Bye (2011)
figs. 6b. and 8c.

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- **What about phonetics?**

§2

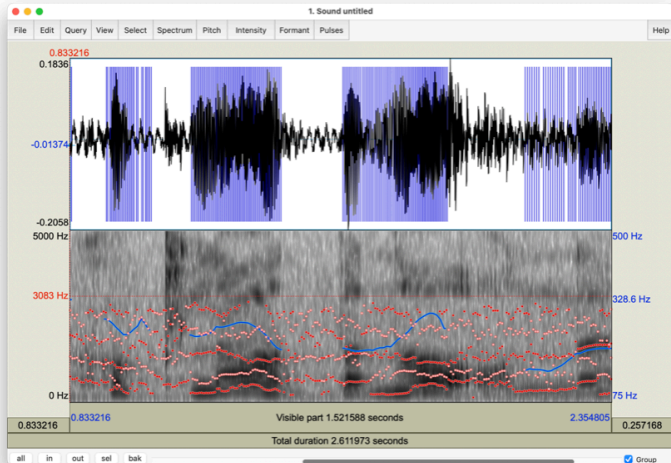
Where does phonological change begin?

The phoneticians' revenge

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“[D]iachrony proposes, synchrony disposes” (Hyman, 2008, 129)

One way to try to adjudicate between them: phonological change is usually *initiated* by phonetic factors, but systemic considerations restrict the set of possible outcomes?

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Phonetics > phonology

- I've made a claim that there are phonetic 'reasons' to think that V[r/ɾ]C sequences and VC_{retroflex} sequences can be perceived similarly. I would like to talk about the grammar. What now?

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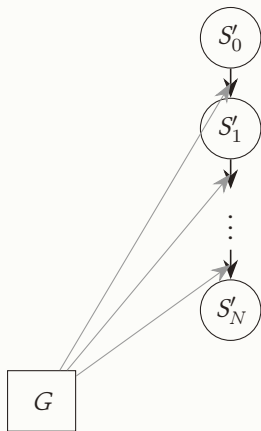
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S'_0

G

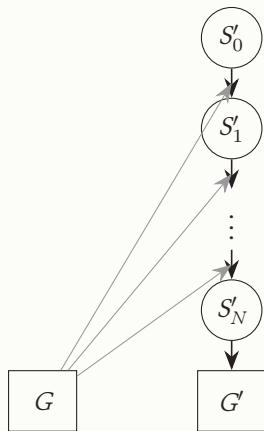
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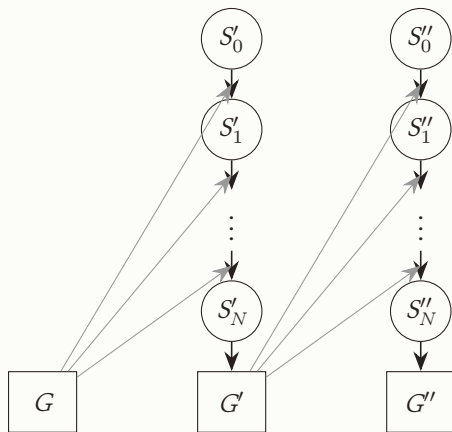
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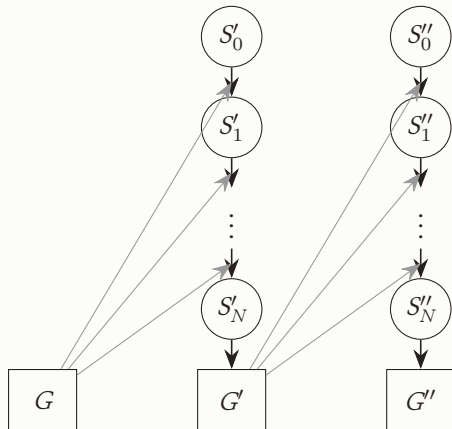
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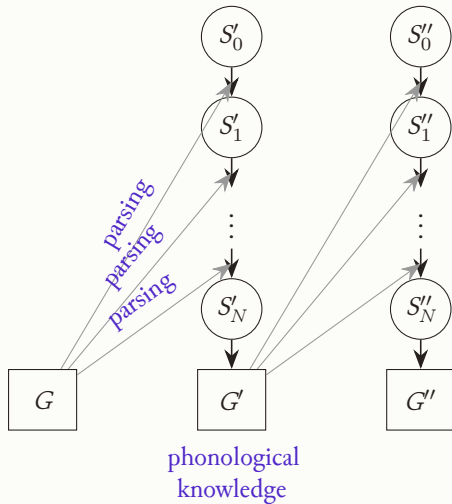
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 - A (the?) basic task that the learner must carry out is *phonological parsing*, i.e. interpret the incoming auditory stimulus as the surface output of a G , and try to work out what that G has to be to generate it.



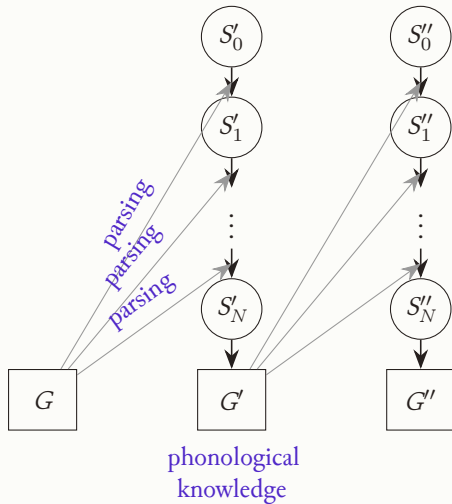
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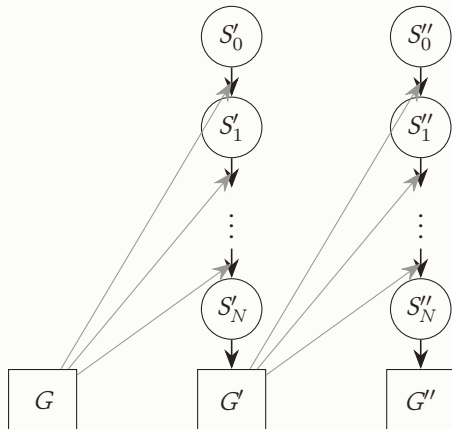
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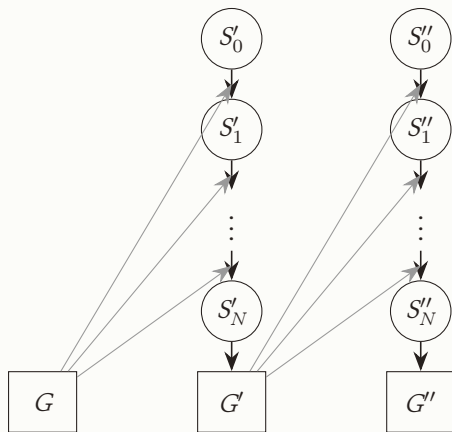
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 - And all they get is noisy data which imperfectly represents G ; a clear locus of *error*. After Ohala (1981, 1989) (...), things can go wrong in various ways: fail to pick up real signal (*hypercorrect*), pick up noise instead (*hypocorrect*) ...



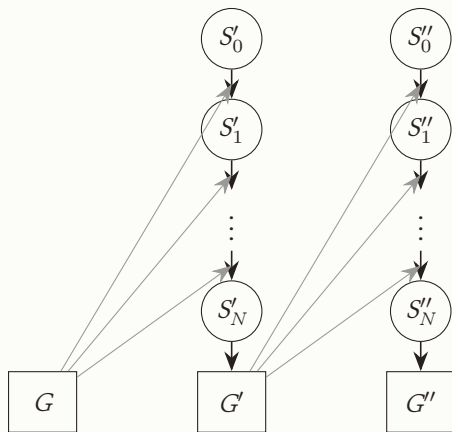
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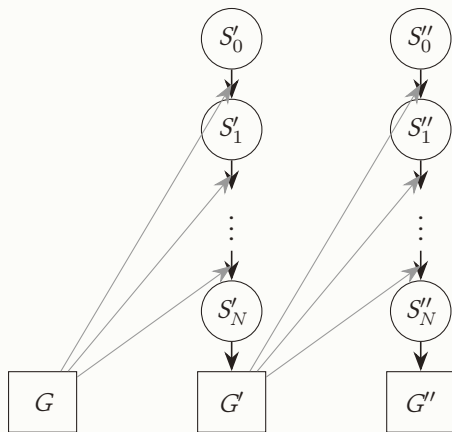
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 - If retroflexion enters the phonology of a variety via perceptual errors driven by low F3, *then we should predict that it does not 'phonologise' readily in varieties with κ .*



Phonologisation and sequels

TODO BRIEF SNAPPY DESCRIPTION OF THE LIFE CYCLE.

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“Retroflexion is a very old process in E. Norwegian, dating back as far as the 14th century (Seip 1955: 177, 289). The Frogner dialect in Oslo is an E. Norwegian dialect [...] Uvular /ɣ/, on the other hand, cannot be very old in the Frogner dialect [...] dating back no further than the middle of the 18th century (Nielsen 1950). Retroflexion must therefore have existed in Oslo long before /r/ was replaced by a uvular /ɣ/ in the Frogner dialect. So when /ɣ/ was introduced into the dialect, the retroflexion process already existed [...]”

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- If we accept this chronology, then we don't need /ɣ/ to *condition* development in Frogner; we just need it to do what we want representationally.

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- *So how does that work?* Stay tuned!

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